Solid State (NCERT)

Time Allowed : 1 Hour ____________________________ **Maximum Marks : 60**

Please read the instructions carefully. You will be alloted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. This question paper consists of 6 questions carrying 10 marks each.

Q1: Classify each of the following solids as ionic, metallic, molecular, network (covalent) or amorphous.

(i) Tetra phosphorus decoxide (P_4O_{10})

(ii) Ammonium phosphate $(NH_4)_3PO_4$

(iii) SiC

(iv) I_2

(v)P_4

(vi) Plastic

(vii) Graphite

(viii) Brass

(ix) Rb

(x) LiBr

(xi) Si

Q2: Silver crystallises in fcc lattice. If edge length of the cell is 4.07×10^{-8} cm and density is 10.5 $g\,cm^{-3}$, calculate the atomic mass of silver.

Q3: Analysis shows that nickel oxide has the formula $Ni_{0.98}O_{1.00}$. What fractions of nickel exist as Ni_{2+} and Ni_{3+} ions?

Q4: Ferric oxide crystallises in a hexagonal close-packed array of oxide ions with two out of every three octahedral holes occupied by ferric ions. Derive the formula of the ferric oxide.

Q5: Explain the following terms with suitable examples:

(i) Schottky defect

(ii) Frenkel defect

(iii) Interstitials and

(iv) F-centres.

Q6: If NaCl is doped with 10^{-3} mol % of $SrCl_2$, what is the concentration of cation vacancies?